Worked Accounting Example: One Complete Quarter

Purpose

This document walks through a single firm's full quarterly accounting cycle, starting from a given balance sheet, applying a set of decisions and market outcomes, and producing the ending balance sheet with every intermediate calculation shown.

This example is the canonical test fixture. When the accounting code is written, it must reproduce these numbers exactly. It is also the specification: any ambiguity in the architecture docs should be resolved by what this example shows.

All values follow the canonical parameters in world doc 09 (Parameters and Calibration).

The Firm: Aeterna Therapeutics (firm_0)

- Quarter: Q2 2031 (the firm's second quarter of operations)
- Mode: Public (post-IPO)
- Generation: Gen 1
- Starting shares outstanding: 10,000,000
- Par value: $0.001 per share
- Capacity: 250 courses/quarter (pilot plant)
- Base unit cost: $14,200 per course (no process R&D effect yet)
- Tax rate: 21%
- Measurement regime: baseline_gaap
- Toggles: entry_exit=ON, financial_institutions=ON, all others OFF

Section 1: Starting Balance Sheet (End of Q1 2031)

These values are the inputs to the Q2 calculation. They are derived from a prior Q1 calculation (shown in the appendix) where the firm IPO'd for $350M, produced 200 courses, sold 180 at $95,000, and invested $25M in R&D and $12M in SGA.

Assets

Line	Amount	Source
Cash (cheq)	303,655,570	IPO 325M minus Q1 cash outflows
Accounts receivable (rectq)	2,565,000	15% of Q1 revenue of 17,100,000
Inventory (invtq)	298,200	20 unsold units from Q1 at 14,910
PPE gross	25,000,000	Pilot plant
Accumulated depreciation	625,000	One quarter of depreciation
PPE net (ppentq)	24,375,000	
Total assets (atq)	330,893,770	

Liabilities

Line	Amount	Source
Accounts payable (apq)	402,570	15% of Q1 COGS of 2,683,800
Accrued expenses (acoq)	3,700,000	10% of Q1 (R&D + SGA) of 37,000,00
Taxes payable (txpq)	0	Q1 was a loss
Current debt (dlcq)	0	No revolver drawn
Long-term debt (dlttq)	0	No term debt
Total liabilities (ltq)	4,102,570	

Equity

Line	Amount	Source
Common stock (cstkq)	10,000	10M shares * $0.001 par
Additional paid-in capital (apicq)	349,990,000	IPO proceeds above par
Retained earnings (req)	(23,208,800)	Q1 net loss
Treasury stock (tstkq)	0	No buybacks
Total equity (ceqq)	326,791,200	

Balance Check

- Total assets: 330,893,770
- Total liabilities + equity: 4,102,570 + 326,791,200 = 330,893,770
- Difference: 0 PASS

Internal State (not on balance sheet)

Variable	Value	Notes
Capability stock (A)	35.0 + 5.0 = 40.0	Start 35 + Q1 R&D effect
Brand stock (B)	10.0 + 1.25 = 11.25	Start 10 + Q1 SGA effect
Cumulative product R&D	10,000,000	From Q1 (60% of 15M discretionary,
Cumulative process R&D	3,750,000	From Q1 (25% of 15M discretionary)
Cumulative delivery R&D	2,250,000	From Q1 (15% of 15M discretionary)
NOL carryforward	23,208,800	Q1 net loss
Product generation	1	
Delivery generation	1	IV infusion

Section 2: Q2 Inputs (Decisions and Outcomes)

Firm Decisions (from firm agent, after clamping)


```
{
  "price": 92000,
  "production": 220,
  "capex": 15000000,
  "rd_spend": 28000000,
  "rd_allocation": {"product": 0.60, "process": 0.25, "delivery": 0.15},
  "sga_spend": 14000000,
  "equity_issuance_request": 0,
  "debt_request": 0,
  "dividends": 0,
  "buybacks": 0
```

```
}
```

Note: rd_spend of 28,000,000 includes the $10M mandatory Phase III trial cost. The discretionary R&D is 28,000,000 - 10,000,000 = 18,000,000, allocated per the above percentages.

Market Outcomes (from environment agent)


```
{
  "firm_0": {
    "units_sold": 200,
    "market_share": 0.22,
    "product_rd_advance": false,
    "process_cogs_reduction_this_quarter": 0.0,
    "delivery_advance": false
  }
}
```


Feasibility Check (Phase 4)

Clamping verification:

- Cash on hand: 303,655,570
- Expected Q2 revenue (80% collected this Q per policy): 200 92,000 0.85 = 15,640,000
- Prior AR collection (all of it): 2,565,000
- Revolver available: 0 (no commitment yet)
- Total available this quarter: 321,860,570

Requested spending:

- COGS cash portion: clamped to actual production cost
- Phase III (mandatory): 10,000,000
- Discretionary R&D: 18,000,000
- SGA: 14,000,000
- Capex: 15,000,000
- Total before COGS: 57,000,000

Available is 321M, requested is well below. No clamping needed. All requested amounts are feasible.

Dividends blocked (RE < 0) -- check passes automatically, dividends=0.

Section 3: Computing the Income Statement

Step 3.1: Effective Unit Cost

```
capacity_utilization = production / capacity
                     = 220 / 250
                     = 0.88
```

Utilization multiplier (from doc 09):

- 0.88 is in the 70-90% band
- Formula: 1.00 + 0.5 * (0.90 - util)
- = 1.00 + 0.5 * (0.90 - 0.88)
- = 1.00 + 0.01

- = 1.01

Process R&D reduction (from cumulative process R&D spending up to start of Q2):

- Cumulative at start of Q2: 3,750,000 (from Q1)
- Formula: 0.22 * (1 - exp(-cumulative / 120,000,000))
- = 0.22 * (1 - exp(-3,750,000 / 120,000,000))
- = 0.22 * (1 - exp(-0.03125))
- = 0.22 * (1 - 0.96923)
- = 0.22 * 0.03077
- = 0.00677 (0.68%)

```
base_cogs_after_process_rd = 14,200 * (1 - 0.00677)
                           = 14,200 * 0.99323
                           = 14,103.87
                           ≈ 14,104

effective_unit_cost = 14,104 * 1.01
                    = 14,245.04
                    ≈ 14,245
```

Step 3.2: Inventory Accounting (FIFO)

Starting inventory: 20 units from Q1 at $14,910 each = 298,200 total.

Q2 production: 220 units at new effective cost of $14,245 each = 3,133,900.

Total units available to sell: 20 (old) + 220 (new) = 240. Units sold: 200. Remaining inventory: 40 units (all from Q2 production).

FIFO cost of goods sold:

- 20 old units * 14,910 = 298,200
- 180 new units * 14,245 = 2,564,100
- COGS = 2,862,300

Ending inventory:

- 40 new units * 14,245 = 569,800

Step 3.3: Income Statement

```
Revenue (saleq)          = 200 * 92,000         = 18,400,000
COGS (cogsq)             =                         2,862,300
Gross profit (gpq)       =                        15,537,700

R&D expense (xrdq)       = 28,000,000
SGA expense (xsgaq)      = 14,000,000
Depreciation (dpq)       = 0.025 * PPE_gross
                         = 0.025 * 25,000,000  =    625,000
                         (capex from Q2 not yet depreciating)

Operating expenses       = 28,000,000 + 14,000,000 + 625,000
                         =                        42,625,000

Operating income (oiadpq) = 15,537,700 - 42,625,000
                         =                      (27,087,300)
```

```
Interest expense (xintq)  =                             0
Pretax income (piq)       =                    (27,087,300)
Tax expense (txtq)        =                             0
                          (loss; no tax; NOL carryforward grows)
Net income (niq)          =                    (27,087,300)
```

Step 3.4: NOL Update

NOL at start of Q2: 23,208,800 (from Q1) Q2 is a loss, so NOL increases:

```
NOL_end_Q2 = 23,208,800 + 27,087,300 = 50,296,100
```

No NOL was used in Q2 (no taxable income to offset).

Section 4: Computing Working Capital Changes

Accounts Receivable

```
Target AR = 15% * current quarter revenue
          = 0.15 * 18,400,000
          = 2,760,000

Collection of prior AR = 100% of opening AR (paid this quarter)
                       = 2,565,000  (becomes cash)

Ending AR = 2,760,000
ΔAR      = 2,760,000 - 2,565,000 = +195,000  (AR grew)
```

Inventory

```
Starting inventory: 298,200
Add production cost: 220 * 14,245 = 3,133,900
Subtract COGS: 2,862,300

Ending inventory = 298,200 + 3,133,900 - 2,862,300 = 569,800
ΔInventory      = 569,800 - 298,200 = +271,600
```

Accounts Payable

```
Target AP = 15% * current quarter COGS
          = 0.15 * 2,862,300
          = 429,345

Payment of prior AP = 100% (paid this quarter)
                    = 402,570  (cash outflow)

Ending AP = 429,345
ΔAP      = 429,345 - 402,570 = +26,775
```

Accrued Expenses

```
Target accrued = 10% * (R&D + SGA)
```

```
               = 0.10 * (28,000,000 + 14,000,000)
               = 4,200,000

Payment of prior accrued = 100%
                         = 3,700,000

Ending accrued = 4,200,000
ΔAccrued      = 4,200,000 - 3,700,000 = +500,000
```

Taxes Payable

```
Ending = 0 (no tax in Q2)
ΔTaxes payable = 0
```

Section 5: Computing the Cash Flow Statement

Cash from Operations (CFO)

```
Net income                            (27,087,300)
+ Depreciation (non-cash)                 625,000
+ ΔAP (source of cash)                     26,775
+ ΔAccrued (source of cash)               500,000
+ ΔTaxes payable                                0
- ΔAR (use of cash)                     (195,000)
- ΔInventory (use of cash)              (271,600)
                                     ------------
CFO (oancfq)                          (26,401,925)
```

Cash from Investing (CFI)

```
- Capex                               (15,000,000)
- Acquisitions                                  0
+ Asset sales                                   0
                                     ------------
CFI (ivncfq)                          (15,000,000)
```

Cash from Financing (CFF)

```
+ Equity issuance                               0
+ New debt                                      0
- Debt repayment                                0
- Dividends                                     0
- Buybacks                                      0
                                     ------------
CFF (fincfq)                                    0
```

Total Change in Cash

```
ΔCash = CFO + CFI + CFF
      = (26,401,925) + (15,000,000) + 0
      = (41,401,925)
```

```
Ending cash = Starting cash + ΔCash
            = 303,655,570 + (41,401,925)
            = 262,253,645
```

Section 6: Computing PPE and Depreciation

```
Starting PPE gross              25,000,000
+ Q2 capex                      15,000,000
Ending PPE gross                40,000,000

Starting accum depreciation        625,000
+ Q2 depreciation                  625,000
                               (on beginning PPE of 25,000,000)
Ending accum depreciation        1,250,000

Ending PPE net  = 40,000,000 - 1,250,000
                = 38,750,000
```

Note: Q2 capex is added to PPE but does not depreciate in Q2 (placed in service at end of quarter -- conservative convention).

Section 7: Computing Retained Earnings

```
Starting RE                    (23,208,800)
+ Net income Q2                (27,087,300)
- Dividends Q2                           0
                               -----------
Ending RE                      (50,296,100)
```

Note: Ending RE equals negative of NOL carryforward by construction (no dividends, no other OCI). This cross-checks the NOL tracking.

Section 8: Computing Internal Stock Updates

Capability Stock (A)

Product R&D in Q2:

- Total R&D: 28,000,000
- Less Phase III: 10,000,000
- Discretionary: 18,000,000
- Product allocation: 60% * 18,000,000 = 10,800,000

```
A_Q2 = (1 - 0.025) * A_Q1 + eta_A * product_rd_spend
     = 0.975 * 40.0 + 0.0008/1,000,000 * 10,800,000
     = 39.0 + 0.00864 * 1,000
     = 39.0 + 8.64
     = 47.64
```

Wait, let me re-check the eta_A formula from doc 09: > eta_A: 0.0008 per $1M

So for $10.8M of product R&D:

```
contribution = 0.0008 * (10,800,000 / 1,000,000)
             = 0.0008 * 10.8
             = 0.00864
```

That gives a tiny increment. Let me re-read the formula: > A_t = (1 - delta_A) A_{t-1} + eta_A actual_product_rd_spend

If eta_A = 0.0008 per $1M, then effectively the formula is:

```
A_t = (1 - 0.025) * A_{t-1} + 0.0008 * (product_rd_spend / 1,000,000)
```

With $10.8M product R&D:

```
A_Q2 = 0.975 * 40.0 + 0.0008 * 10.8
     = 39.0 + 0.00864
     = 39.00864
```

That's essentially no growth. Doc 09 says "$30M/quarter -> +24 points/quarter (net of depreciation)". Let me check:

With $30M product R&D:

```
A_growth = 0.0008 * 30 = 0.024  (NOT 24)
```

There's a unit mismatch in doc 09. The intended behavior is probably:

```
eta_A * product_rd_spend_in_millions = 0.8 per $1M
```

So with $30M -> 0.8 * 30 = 24 points gross. After depreciation:

```
A_new = 0.975 * A_old + 0.8 * 30
      = 0.975 * A_old + 24
```

CORRECTION TO DOC 09 NEEDED: eta_A should be 0.8 per $1M (not 0.0008 per $1M). Same issue for eta_B.

Using corrected eta_A = 0.8 per $1M:

```
A_Q2 = 0.975 * 40.0 + 0.8 * 10.8
     = 39.0 + 8.64
     = 47.64
```

Brand Stock (B)

SGA spend Q2: 14,000,000. Effective quality at start of Q2 (approximate): ~35 (Gen 1 baseline). Quality effectiveness factor: 35 / 50 = 0.70.

Using corrected eta_B = 1.5 per $1M:

```
B_Q2 = (1 - 0.10) * B_Q1 + eta_B * sga_millions * quality_effectiveness
     = 0.90 * 11.25 + 1.5 * 14.0 * 0.70
     = 10.125 + 14.70
     = 24.825
```

R&D Cumulative Tracking

```
Cumulative product R&D = 10,000,000 + 10,800,000 = 20,800,000
Cumulative process R&D =  3,750,000 +  4,500,000 =  8,250,000
Cumulative delivery R&D=  2,250,000 +  2,700,000 =  4,950,000

(Q2 discretionary: 18,000,000 * 0.25 = 4,500,000 process; 18,000,000 * 0.15 = 2,700,000 delivery)
```

None of these have crossed the Gen 2 threshold ($400M minimum), so no stochastic advance check runs yet.

Capacity Stock

Starting capacity: 250 (pilot plant) Q2 capex: 15,000,000 -- this is below the threshold for any meaningful new facility. Treat as maintenance capex (capacity stays at 250).

Alternatively, treat it as the start of a multi-quarter build (doc 03 small commercial plant: $120M over 4 quarters = $30M/Q). 15M is below that rate too.

For this example: capex is maintenance + pilot augmentation. Capacity stays 250. The 15M is added to PPE gross but does not change capacity.

Section 9: Ending Balance Sheet

Assets

Line	Amount	Derivation
Cash	262,253,645	303,655,570 + (-41,401,925)
Accounts receivable	2,760,000	15% of Q2 revenue
Inventory	569,800	40 units * 14,245
PPE gross	40,000,000	25M + 15M capex
Accum depreciation	(1,250,000)	625K + 625K
PPE net	38,750,000	
Total assets	304,333,445	

Liabilities

Line	Amount	Derivation
Accounts payable	429,345	15% of Q2 COGS
Accrued expenses	4,200,000	10% of (R&D + SGA)
Taxes payable	0	Q2 loss
Current debt	0	
Long-term debt	0	
Total liabilities	4,629,345	

Equity

Line	Amount	Derivation
Common stock	10,000	Unchanged
APIC	349,990,000	Unchanged
Retained earnings	(50,296,100)	(23,208,800) + (27,087,300)
Treasury stock	0	Unchanged
Total equity	299,703,900	

Section 10: Invariant Verification

Invariant 1: Balance Sheet Identity

```
Total assets = 304,333,445
Total liabilities + equity = 4,629,345 + 299,703,900 = 304,333,245
Difference = 304,333,445 - 304,333,245 = 200
```

FAIL by $200 -- this is a rounding issue to fix. Let me find it.

Tracing: the mismatch is exactly 200. Checking RE:

- Starting RE: (23,208,800)
- Net income: (27,087,300)
- Expected ending RE: (50,296,100)
- Recomputing: 23,208,800 + 27,087,300 = 50,296,100

Checking cash:

- Start: 303,655,570
- CFO: (26,401,925)
- CFI: (15,000,000)
- End: 262,253,645
- Check: 303,655,570 - 26,401,925 - 15,000,000 = 262,253,645

Checking total assets arithmetic:

```
262,253,645 + 2,760,000 + 569,800 + 38,750,000
= 264,013,645 + 569,800 + 38,750,000
= 264,583,445 + 38,750,000
= 303,333,445
```

Hmm I wrote 304 earlier. Recounting:

```
262,253,645
+ 2,760,000  -> 265,013,645
+    569,800 -> 265,583,445
+ 38,750,000 -> 304,333,445
```

That's 304,333,445.

Total L+E:

```
  4,629,345
+ 10,000     -> 4,639,345
+ 349,990,000 -> 354,629,345
- 50,296,100 -> 304,333,245
```

Hmm so 304,333,245 vs 304,333,445. Difference is exactly 200.

Let me recheck inventory. Q2 production 220 at 14,245:

```
220 * 14,245 = ?
220 * 14,000 = 3,080,000
220 * 245 = 53,900
Total = 3,133,900
```

40 units * 14,245 = 569,800

Hmm, 14,245 40 = 569,800. Let me verify: 14,245 4 = 56,980. * 10 = 569,800.

COGS: FIFO

```
20 units at 14,910 = 298,200
180 units at 14,245 = ?
180 * 14,245 = 180 * 14,000 + 180 * 245
             = 2,520,000 + 44,100
             = 2,564,100
Total COGS = 298,200 + 2,564,100 = 2,862,300
```

Now check inventory continuity:

```
Start inv: 298,200
Add production: 3,133,900
Less COGS: 2,862,300
End inv: 298,200 + 3,133,900 - 2,862,300

298,200 + 3,133,900 = 3,432,100
3,432,100 - 2,862,300 = 569,800
```

OK that's right.

Check CFO:

```
NI:            (27,087,300)
+ Dep:              625,000
+ ΔAP:               26,775
+ ΔAcc:             500,000
- ΔAR:             (195,000)
- ΔInv:            (271,600)
```

Wait: ΔInv should be 569,800 - 298,200 = 271,600

CFO sum:

```
-27,087,300 + 625,000 = -26,462,300
-26,462,300 + 26,775 = -26,435,525
-26,435,525 + 500,000 = -25,935,525
-25,935,525 - 195,000 = -26,130,525
-26,130,525 - 271,600 = -26,402,125
```

Hmm I got (26,402,125), not (26,401,925). Off by 200.

Let me recompute ΔAR:

```
Target AR = 0.15 * 18,400,000 = 2,760,000
Start AR = 2,565,000
ΔAR = 2,760,000 - 2,565,000 = 195,000
```

And the accounting treatment: during the quarter, 100% of prior AR is collected, 85% of new revenue collected, 15% becomes new AR. Let me verify via direct cash:

Cash inflow from revenue + AR:

```
Revenue collected: 0.85 * 18,400,000 = 15,640,000
Prior AR collected: 2,565,000
Total cash in: 18,205,000
```

New AR at end of Q2 = 0.15 * 18,400,000 = 2,760,000

The direct CFO method (more intuitive):

```
Cash from customers = 18,205,000
- Cash paid to suppliers (COGS - ΔAP) = 2,862,300 - 26,775 = 2,835,525
- Cash paid for R&D (full, since accrued is 10% * combined)
  Actually: R&D + SGA = 42,000,000, ending accrued = 4,200,000, starting accrued = 3,700,000
  Cash paid = 42,000,000 - (4,200,000 - 3,700,000) = 42,000,000 - 500,000 = 41,500,000
- Taxes paid = 0
- Interest paid = 0

CFO = 18,205,000 - 2,835,525 - 41,500,000
```

```
    = 18,205,000 - 44,335,525
    = (26,130,525)
```

So the direct method gives CFO = (26,130,525).

The indirect method gave (26,402,125). These disagree by 271,600 which is exactly the inventory change!

Found the bug. In the indirect method, I subtracted the 271,600 inventory change, but inventory is a NON-CASH build. Production cost of 3,133,900 came from where? Raw materials purchased (becoming AP then cash out) plus labor (becoming accrued then cash out). The inventory build represents cost capitalized, not yet expensed.

The correct indirect CFO:

```
NI                      (27,087,300)
+ Dep                       625,000
Changes in WC:
+ ΔAP                        26,775
+ ΔAccrued                  500,000
- ΔAR                      (195,000)
- ΔInventory               (271,600)

Sum: (26,402,125)
```

But the direct method shows (26,130,525). Discrepancy = 271,600.

The issue is that in the indirect method, when you subtract ΔInventory, you are saying "we spent 271,600 building inventory (cash out)". But production cost INCLUDES expenses that were already NON-CASH (accrued).

Actually no. Let me think more carefully.

The subtlety: Cost of production = 3,133,900. This consists of:

- Materials (cash or AP)
- Labor (cash or accrued)
- Overhead (depreciation, utilities)

Of this 3,133,900:

- 2,862,300 becomes COGS (expensed)
- 271,600 becomes inventory (capitalized)

When we compute NI, we subtract only the 2,862,300 (the COGS portion). The 271,600 is NOT expensed, so it doesn't reduce NI.

But the 271,600 WAS spent (cash or accrual). If it was cash, we need to subtract it from CFO. If it was accrued, we don't.

The issue is that my working capital model doesn't separate "production costs unpaid" from "production costs paid as cash". In the simplified model with theta_AP = 15% of COGS, AP relates only to expensed COGS, not to inventory build.

To fix this: either (a) treat 100% of production as cash (no AP), or (b) AP covers full production cost (not just COGS).

Cleanest fix: AP ratio is applied to total production cost, not COGS. So:

```
AP = 0.15 * (production * unit_cost)
   = 0.15 * 3,133,900
   = 470,085
```

Let me redo with this convention:

```
Target AP = 0.15 * 3,133,900 = 470,085
ΔAP = 470,085 - 402,570 = 67,515
```

Payment flow:

- Pay prior AP: 402,570 (cash out)
- Incur new AP: 470,085 (deferred)
- Net AP payment this Q: 402,570 - 0.15 * prior_production_cost...

Actually wait, this is getting complicated. Let me use a different, cleaner convention: Production cost is ALL paid in cash (no AP). Inventory is financed by cash. Then AP applies only to COGS-related purchases during the period, which is effectively the same as the old theta_AP = 15% of COGS.

Under this cleaner rule:

- Cost of production 3,133,900 is entirely cash outflow
- AP ratio becomes a lag on COGS-specific vendor payments, which is a separate

abstraction. For simplicity, let's REMOVE AP and assume cash-basis supplier payments.

But that changes the whole model. Let me try yet another approach that keeps the accrual model.

Alternative fix: separate the inventory-building cash from the COGS-related cash.

Direct method breakdown:

```
Cash to suppliers for COGS-related materials = COGS - ΔAP (AP applies to COGS)
                                              = 2,862,300 - 26,775
                                              = 2,835,525

Cash to build inventory (not yet in COGS) = production_cost - COGS
                                          = 3,133,900 - 2,862,300
                                          = 271,600
```

Total cash to suppliers = 2,835,525 + 271,600 = 3,107,125

Then:

```
CFO direct =
  Cash from revenue + AR:    18,205,000
  - Cash to suppliers:        (3,107,125)
  - Cash for R&D+SGA:        (41,500,000)
  - Interest:                         0
  - Taxes:                            0
  = CFO:                     (26,402,125)
```

This matches the indirect method. Good.

So the actual CFO is (26,402,125), not (26,130,525). My direct method above missed the inventory build cost.

Let me recompute ending cash:

```
Starting cash: 303,655,570
+ CFO:         (26,402,125)
+ CFI:         (15,000,000)
+ CFF:                   0
= Ending cash: 262,253,445
```

That's 262,253,445 (not 262,253,645 as I had before). The 200 discrepancy was in my arithmetic.

Let me re-verify:

```
303,655,570
- 26,402,125 = ?

303,655,570
-26,402,125
```

```
----------
277,253,445  (303M - 26M = 277M, then adjustments)

Actually: 303,655,570 - 26,402,125
= 303,655,570 - 26,000,000 - 402,125
= 277,655,570 - 402,125
= 277,253,445

Then: 277,253,445 - 15,000,000 = 262,253,445
```

Ending cash = 262,253,445.

Recomputing total assets:

```
262,253,445 + 2,760,000 + 569,800 + 38,750,000
```

```
262,253,445 + 2,760,000 = 265,013,445
265,013,445 + 569,800 = 265,583,245
265,583,245 + 38,750,000 = 304,333,245
```

Total assets = 304,333,245.

Total L+E was 304,333,245 (computed earlier).

BALANCE SHEET IDENTITY: 304,333,245 = 304,333,245 PASS

The original $200 error was an arithmetic slip in cash computation. With the corrected cash of 262,253,445 (not ...645), everything balances.

Invariant 2: Cash Reconciliation

```
ΔCash = CFO + CFI + CFF
      = (26,402,125) + (15,000,000) + 0
      = (41,402,125)

Ending cash - starting cash = 262,253,445 - 303,655,570 = (41,402,125)
```

Invariant 3: Retained Earnings Roll-Forward

```
RE_end = RE_start + NI - dividends + other_comprehensive_income
       = (23,208,800) + (27,087,300) - 0 + 0
       = (50,296,100)
```

Invariant 4: Inventory Continuity

```
End inv = Start inv + production_cost - COGS
        = 298,200 + 3,133,900 - 2,862,300
        = 569,800
```

Invariant 5: PPE Continuity

```
End PPE gross = Start PPE gross + capex
              = 25,000,000 + 15,000,000
              = 40,000,000

End accum dep = Start accum dep + Q2 dep
              = 625,000 + 625,000
              = 1,250,000
```

Invariant 6: Non-Negative Cash

```
Ending cash = 262,253,445 > 0
```

Invariant 7: Non-Negative Total Assets

```
atq = 304,333,245 > 0
```

Section 11: Final Ending State Snapshot

```
Firm: firm_0
Quarter: Q2 2031
Status: active

BALANCE SHEET
  Cash:                        262,253,445
  Accounts receivable:           2,760,000
  Inventory:                       569,800
  PPE gross:                    40,000,000
  Accum depreciation:          (1,250,000)
  PPE net:                      38,750,000
  TOTAL ASSETS                 304,333,245

  Accounts payable:                429,345
  Accrued expenses:              4,200,000
  Taxes payable:                         0
  Current debt:                          0
  Long-term debt:                        0
  TOTAL LIABILITIES              4,629,345

  Common stock:                     10,000
  APIC:                        349,990,000
  Retained earnings:          (50,296,100)
  Treasury stock:                        0
  TOTAL EQUITY                 299,703,900

  LIAB + EQUITY                304,333,245

INCOME STATEMENT
  Revenue:                      18,400,000
  COGS:                          2,862,300
  Gross profit:                 15,537,700
  R&D expense:                  28,000,000
  SGA expense:                  14,000,000
  Depreciation:                    625,000
  Operating income:           (27,087,300)
  Interest:                              0
  Pretax income:              (27,087,300)
  Tax expense:                           0
  NET INCOME                  (27,087,300)
```

```
CASH FLOW STATEMENT
  CFO:                        (26,402,125)
  CFI:                        (15,000,000)
  CFF:                                   0
  CHANGE IN CASH              (41,402,125)

INTERNAL STATE
  Capability stock (A):              47.64
  Brand stock (B):                   24.83
  Cumulative product R&D:       20,800,000
  Cumulative process R&D:        8,250,000
  Cumulative delivery R&D:       4,950,000
  NOL carryforward:             50,296,100
  Product generation:                    1
  Capacity:                            250
  Effective unit cost:              14,245
```

Section 12: Notes for the Implementation

Issues Found During This Exercise

• Doc 09 parameter bug: eta_A and eta_B in doc 09 are stated as "0.0008 per $1M" and "0.0015 per $1M" but the example in doc 09 says "$30M/quarter -> +24 points". The correct values are eta_A = 0.8 per $1M and eta_B = 1.5 per $1M. Doc 09 needs correction.

• Working capital timing: The convention used here is:

- AR = theta_AR * current quarter revenue (NOT including prior AR)
- Prior period AR is fully collected each quarter (100% turnover)
- AP = theta_AP * current quarter COGS (NOT production cost)
- Prior period AP is fully paid each quarter
- Accrued = theta_accrued * (R&D + SGA) of current quarter
- Prior period accrued is fully settled each quarter

• Inventory build as non-cash: Production cost exceeds COGS by the inventory build amount (271,600 in this example). This must be treated as a cash outflow in the CFO statement (the cost was incurred but not expensed). The indirect method does this automatically via the ΔInventory adjustment. The direct method must separately account for the inventory build cash.

• Depreciation timing: Q2 capex is added to PPE gross but does not generate Q2 depreciation (placed in service at end of quarter, depreciates starting Q3). This is a conservative convention.

• Inventory FIFO: Old inventory sold first at old cost; new units sold at new cost. This matches standard FIFO accounting.

• NOL tracking: Retained earnings and cumulative NOL should equal each other (negated) as long as no dividends and no OCI. This is a useful cross-check.

What This Example Does NOT Cover

• Stock-based compensation (toggle OFF)

• Lease accounting (toggle OFF)

- Goodwill / M&A (toggle OFF)
- Restructuring charges (no layoffs)
- R&D capitalization (baseline GAAP regime)
- Fair value adjustments (baseline GAAP regime)
- Diluted shares (no options/warrants)
- Process R&D generation advance (below threshold)
- Interest expense (no debt)
- Tax expense (pretax loss)
- Revolver draws (not needed)

Each of these should get its own worked example as the corresponding feature is implemented.

Section 13: Test Fixture (for pytest)

```
# tests/fixtures/q2_2031_firm_0.py

STARTING_STATE = FirmState(
    firm_id="firm_0",
    quarter=1,  # END of Q1
    cash=303_655_570,
    accounts_receivable=2_565_000,
    inventory_units=20,
    inventory_value=298_200,
    ppe_gross=25_000_000,
    accum_depreciation=625_000,
    accounts_payable=402_570,
    accrued_expenses=3_700_000,
    taxes_payable=0,
    revolver_balance=0,
    long_term_debt=0,
    common_stock=10_000,
    apic=349_990_000,
    retained_earnings=-23_208_800,
    treasury_stock=0,
    shares_outstanding=10_000_000,
    capability_stock=40.0,
    brand_stock=11.25,
    rd_cumulative_product=10_000_000,
    rd_cumulative_process=3_750_000,
    rd_cumulative_delivery=2_250_000,
    nol_carryforward=23_208_800,
    product_generation=1,
    capacity_units=250,
    base_unit_cost=14_200,
)

Q2_DECISIONS = Decisions(
    price=92_000,
    production=220,
    capex=15_000_000,
```

```
    rd_spend=28_000_000,
    rd_allocation={"product": 0.60, "process": 0.25, "delivery": 0.15},
    sga_spend=14_000_000,
    equity_issuance_request=0,
    debt_request=0,
    dividends=0,
    buybacks=0,
)

Q2_OUTCOMES = MarketOutcomes(
    firm_id="firm_0",
    units_sold=200,
    market_share=0.22,
    product_rd_advance=False,
    process_cogs_reduction_this_quarter=0.0,
    delivery_advance=False,
)

EXPECTED_END_STATE = FirmState(
    firm_id="firm_0",
    quarter=2,
    cash=262_253_445,
    accounts_receivable=2_760_000,
    inventory_units=40,
    inventory_value=569_800,
    ppe_gross=40_000_000,
    accum_depreciation=1_250_000,
    accounts_payable=429_345,
    accrued_expenses=4_200_000,
    taxes_payable=0,
    revolver_balance=0,
    long_term_debt=0,
    common_stock=10_000,
    apic=349_990_000,
    retained_earnings=-50_296_100,
    treasury_stock=0,
    shares_outstanding=10_000_000,
    capability_stock=47.64,
    brand_stock=24.825,
    rd_cumulative_product=20_800_000,
    rd_cumulative_process=8_250_000,
    rd_cumulative_delivery=4_950_000,
    nol_carryforward=50_296_100,
    product_generation=1,
    capacity_units=250,
    base_unit_cost=14_200,
)

EXPECTED_FLOWS = QuarterFlows(
    # Income statement
    net_sales=18_400_000,
```

```
    cogs=2_862_300,
    gross_profit=15_537_700,
    rd_expense=28_000_000,
    sga_expense=14_000_000,
    depreciation=625_000,
    operating_income=-27_087_300,
    interest_expense=0,
    pretax_income=-27_087_300,
    tax_expense=0,
    net_income=-27_087_300,
    # Cash flow
    cfo=-26_402_125,
    cfi=-15_000_000,
    cff=0,
    change_in_cash=-41_402_125,
    # Actuals (after clamping -- no clamping in this example)
    actual_price=92_000,
    actual_production=220,
    actual_capex=15_000_000,
    actual_rd_spend=28_000_000,
    actual_sga_spend=14_000_000,
    units_sold=200,
    market_share=0.22,
)


def test_q2_2031_posting():
    result_state, result_flows = post_quarter(
        prior_state=STARTING_STATE,
        decisions=Q2_DECISIONS,
        outcomes=Q2_OUTCOMES,
        params=DEFAULT_PARAMS,
    )

    assert_state_equal(result_state, EXPECTED_END_STATE, tol=1.0)
    assert_flows_equal(result_flows, EXPECTED_FLOWS, tol=1.0)

    # Invariants
    assert abs(result_state.total_assets - result_state.total_liabilities - result_state.total_equity)
    assert result_state.cash >= 0
    assert result_state.total_assets >= 0
    assert result_state.retained_earnings == STARTING_STATE.retained_earnings + result_flows.net_incom
    assert result_flows.change_in_cash == result_flows.cfo + result_flows.cfi + result_flows.cff
```

This fixture becomes the first regression test. Any change to the accounting module must produce these exact numbers (within $1 tolerance for rounding).